

The paradox of `itertools.tee`

PyCon Lithuania 🇱🇹



Rodrigo 🐍 🚀

mathspp.com/insider



```
squares = (x ** 2 for x in range(3))  
  
print(squares)
```



```
squares = (x ** 2 for x in range(3))  
  
print(squares)
```

```
<generator object <genexpr> at ... >
```




```
squares = (x ** 2 for x in range(3))  
  
print(list(squares))
```




```
squares = (x ** 2 for x in range(3))  
  
print(list(squares))
```

```
[0, 1, 4]
```



```
squares = (x ** 2 for x in range(3))  
  
print(1 in squares)
```



```
squares = (x ** 2 for x in range(3))  
  
print(1 in squares)
```

True



```
squares = (x ** 2 for x in range(3))
```

```
print(1 in squares)
```

```
print(1 in squares)
```



```
squares = (x ** 2 for x in range(3))
```

```
print(1 in squares)
```

```
print(1 in squares)
```

True

False

Generators are **iterators**.

Generators are **iterators**.

We usually think in terms of **iterables**.

Iterables vs iterators

Iterable: something you can iterate over.

Iterable: something you can iterate over.

vs

Iterator: low-level object that knows
how to iterate.

List

List

vs

- `list_iterator`

List

vs

- `list_iterator`
- `list_reverseiterator`

Iterator laziness comes at a cost

Iterator laziness comes at a cost



Iterators are less flexible

Iterator laziness comes at a cost



Iterators are less flexible



Iterators can only be traversed *once*

Iterator laziness comes at a cost



Iterators are less flexible



Iterators don't have a length

Iterator laziness comes at a cost



Iterators are less flexible



Iterators can't be indexed

Iterator laziness comes at a cost



Iterators are less flexible



Iterators can't be sliced

itertools ❤ iterators

iter-tools

itertools menu

CONTENT WARNING



accumulate	cycle	permutations
batched	dropwhile	product
chain	filterfalse	repeat
combinations	groupby	starmap
combinations_with_replacement		takewhile
compress	islice	tee
count	pairwise	zip_longest

Filtering

compress
dropwhile
filterfalse
takewhile

Complementary

accumulate
starmap
zip_longest

Reshaping

batched
chain
groupby
islice
pairwise

Combinatorial

permutations
product
combinations
combinations_with_replacement

Infinite

count
cycle
repeat

tee

Filtering

compress
dropwhile
filterfalse
takewhile

Complementary

accumulate
starmap
zip_longest

Reshaping

batched
chain
groupby
islice
pairwise

Combinatorial

permutations
product
combinations
combinations_with_replacement

Infinite

count
cycle
repeat



Live ~~bugs~~demo

Summary

- tee is low-level

- tee is low-level
- allows you to go against iterator rules

- tee is low-level
- allows you to go against iterator rules
- but it's all smoke and mirrors

Understanding the inner workings of Python

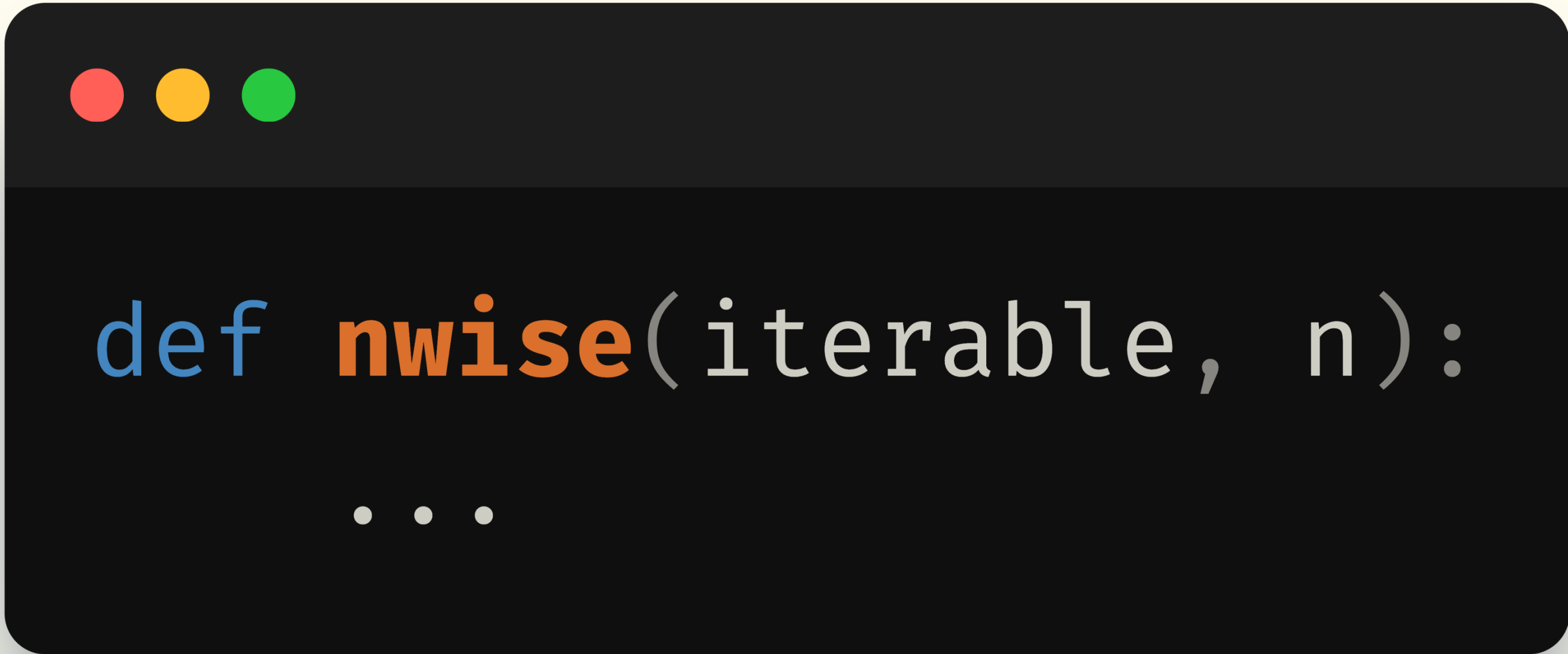
Understanding the inner
workings of Python



Freedom and flexibility

Challenge

Implement `nwise`,
a generalisation of `pairwise`



```
def nwise(iterable, n):  
    ...
```




Rodrigo 🐍 🚀

mathspp.com/insider

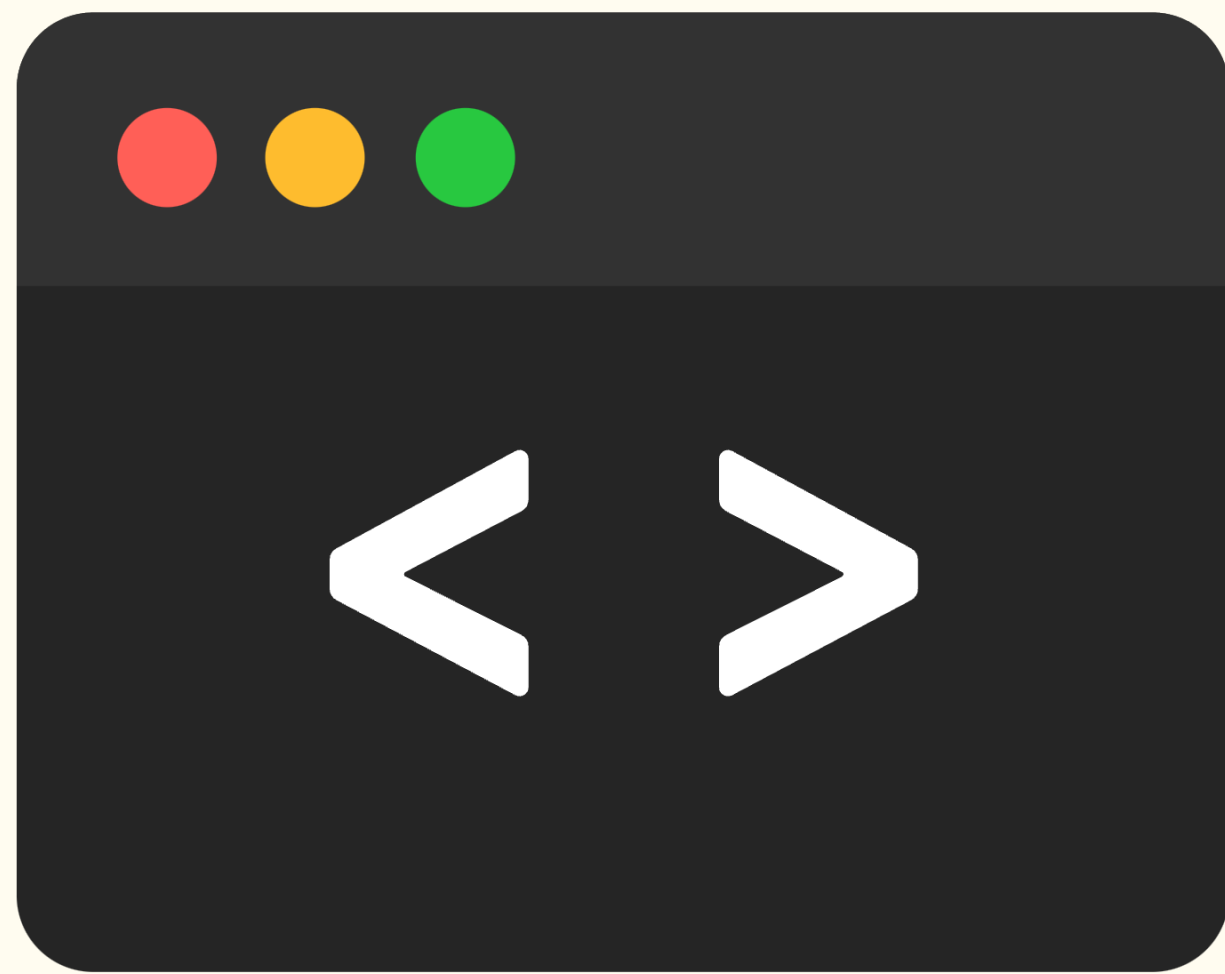
Become the
smartest
Python dev in
the room 🐍🚀



Bonus slides ✨

Session notebook





snappify